Daniel Gabriele

Software Engineer — ML Research Engineer

New York City, NY

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Professional Summary

Engineer with 12+ years across full-stack product development, web3 infrastructure, high-performance computing, and ML research. I've shipped production systems at every layer—microservices and fintech platforms, CosmWasm smart contracts and on-chain architectures, real-time trading systems running on live capital—and I build research software independently, from neural operator pre-training frameworks to multilingual VAE decoders for emergent agent communication.

My earlier work in HPC (non-blocking data structures, parallel algorithms) and scientific computing still shapes how I think about systems at any level of the stack. Current interests center on quantitative trading and systematic strategy development, and on AI research in pre-training methods and scientific machine learning—neural operators, emergent communication, and foundation models for physical systems.

I came to computer science through comparative religion and a curiosity about complex systems, not through a traditional engineering track. That path has had a lasting influence on how I approach design problems—I tend to reach for ideas across disciplines and pick up new domains quickly.

Work Experience

Independent Algorithmic Trader & Engineer

Jan 2025 - Present — Self-Employed

- Built and operate a production trading system across multiple decentralized exchanges, with a real-time data pipeline processing live market data streams
- Created an open source genetic programming framework (Nature) for evolving adaptive trading rule ensembles with self-regulating behavior
- Trained gradient boosting metafilters for orderbook dynamics prediction, with genetic programming handling automated feature engineering
- Designed the full real-time stack for deploying, monitoring, and adapting strategies with minimal latency
- **Technologies:** Python, FastAPI, WebSockets, PyTorch, Pandas, NumPy, Scipy

Co-Founder & Technical Lead — Gelotto

Feb 2022 - Nov 2025 — Self-Employed

Web3 company building decentralized games, social platforms, and DeFi applications across multiple Cosmos networks, with validators and open source infrastructure contributions.

- Wrote 20+ production CosmWasm smart contracts powering games, social platforms, and DeFi protocols across 6 Cosmos networks
- Developed a "zero-backend" architecture—complete applications living entirely on-chain, with no hosting costs or private RPC dependencies
- Launched pampit.fun, a token launchpad powered 100% by smart contracts and handling real transaction volume
- Built Threads, a Reddit-like fully on-chain social app with tree-structured discussions and native tipping
- Created multiple lottery and gaming platforms with profit-sharing mechanisms
- Launched three NFT series with trustless on-chain profit distribution
- Won governance approval to deploy on five networks; received multiple grants; reached $400K peak LP investment during ICO
- Grew and managed a large Discord community; presented at in-person Cosmos events
- **Technologies:** Rust, CosmWasm, TypeScript, Next.js, Nest.js, Expo, Docker, Kubernetes

Senior Software Engineer — Strike Protocols

Jan 2021 - Dec 2022 — New York City, NY — Crunchbase

Fintech startup building institutional-grade custody and capital markets infrastructure for digital assets.

- Built multi-stage threshold authorization for a Solana enterprise vault application
- Designed frontend workflows and vault management UI/UX for enterprise clients
- Contributed to the async backend framework syncing on-chain state with internal databases
- **Technologies:** Rust, TypeScript, Kotlin, Solana

Founder & Lead Engineer — Artfactory

Mar 2021 - Oct 2022 — Self-Employed

- Built a full-stack dropshipping platform around a custom procedural art generation library (Reactor)
- Developed parametric art engine rendering high-resolution outputs onto product mockups on demand
- Designed the backend with async task processing for on-the-fly generation and rendering
- **Technologies:** Node.js, TypeScript, Canvas API, Python, Celery

Senior Software Engineer (Contractor) — Private Investment Firm

Jan 2019 - Dec 2021 — Contract

- Developed automated trading system for private investor with real-time market data streaming pipeline
- Designed multiscale adaptive trading strategies according to client specifications
- Built backend infrastructure for storing and processing high-frequency market data
- **Technologies:** Python, FastAPI, WebSockets, Pandas, NumPy, Scipy

Senior Software Engineer — Axial

Feb 2014 - Nov 2019 — New York, NY

Private deal network connecting middle-market business owners, M&A advisors, and investors through proprietary matching algorithms.

- Led migration from monolith to FastAPI-based microservices for the core deal platform
- Designed and built the matching engine connecting business owners with investors and buyers
- Built search and recommendation systems for discovering investment opportunities
- Implemented authentication and authorization across the platform
- **Technologies:** Python, FastAPI, Celery, PostgreSQL, Redis, Docker, Kubernetes

Independent Android Developer

Feb 2013 - Nov 2014 — Self-Employed

- Built five Android applications including form builders, P2P commerce, budgeting, and collaboration tools
- Handled full development lifecycle from design to Play Store deployment

Research Assistant — University of Central Florida

2009 - 2013

- Developed non-blocking data structures for HPC applications (hash sets, bloom filters, green threads) in C++
- Built multitouch applications exploring symbolic learning and emergent communication via cellular automata

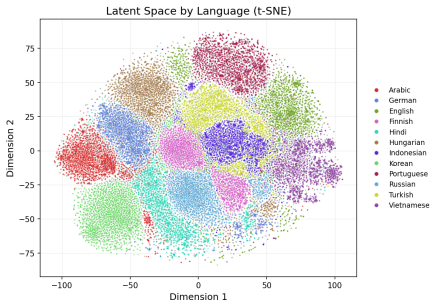
- Created algebra learning games for an experimental high school curriculum
- Contributed to evolutionary computing research on adaptive systems
- **Technologies:** C++, Python, X Window System

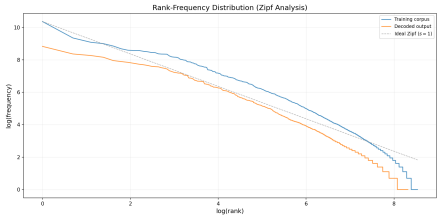
Selected Technical Projects

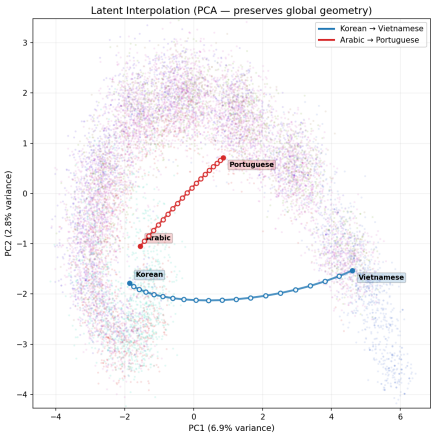
LFM — Language Faculty Model

GitHub: github.com/dgabriele/lfm — PyTorch, VAE, PonderNet, SentencePiece

Agents embedded in complex systems develop internal representations that encode perspectives no human has access to—but have no way to express them. LFM gives those agents a language. A multilingual VAE decoder pretrained on 4M leaf-level phrase constituents from 12 typologically diverse languages is frozen as a linguistic bottleneck, analogous to a Chomskyan Universal Grammar: agents learn only to project representations into the decoder's latent space, and the fixed structural prior constrains output to be phonotactically well-formed, compositional, and variable-length. A GRU z-sequence generator with PonderNet halting produces multi-segment expressions through the frozen decoder, with data-driven z diversity regularization that prevents segment collapse while maintaining linguistic coherence. Achieves 97.7% accuracy (15.6x above chance) on 16-way referential games with 100% hard semantic negatives. Latent space recovers typological groupings across language families, exhibits near-perfect Zipfian statistics (corpus 1.004, decoded 1.058), adaptive length scaling (r=1.000), and smooth cross-typological interpolation. Includes CLI visualization suite and expression sampling tools.







Spinlock — Neural Operator Pre-Training Framework

GitHub: github.com/dgabriele/spinlock — PyTorch, U-AFNO, Neural Operators, VQ-VAE

Pre-training framework for neural operators over continuous dynamical systems (e.g. Lenia continuous cellular automata, reaction-diffusion PDEs). Generates large-scale trajectory datasets via Sobol-stratified parameter sampling, extracts multi-modal behavioral features, and trains U-AFNO meta-operators as fast surrogate world models across diverse dynamics. Includes two tokenization pathways: a VQ-VAE discrete codebook for behavioral clustering, and an NLTokenizer that integrates with LFM's frozen linguistic decoder—encoding trajectory dynamics into a continuous VAE latent space, projecting through Gumbel-Softmax into LFM's frozen autoregressive decoder, and verifying fidelity via a listener roundtrip loss. The goal: agents that observe dynamical systems and describe what they see in their own emergent language, translatable to English via a fine-tuned LLM.

Nature — Genetic Programming Framework

GitHub: github.com/dgabriele/nature — Python, Grammar Evolution

Open source framework that evolves Python instance methods via `@evolve` decorator. Multi-objective optimization over arbitrary datatypes (Pandas, NumPy, CuPy) outperforming DEAP. Built for generating adaptive trading rule ensembles.

Reactor — Procedural Generative Art Framework

GitHub: github.com/superginyuforce/reactor — Node.js, TypeScript, Canvas API

Framework for procedural art generation with unified brush abstraction supporting parametric control over geometric primitives. Implements lazy evaluation proxy pattern for stroke dynamics with deterministic PRNG for reproducible variation. Produces high-resolution outputs (15000×15000px) for physical mockups and print-on-demand.

PampIt — Token Launchpad

GitHub: github.com/Gelotto/pampit — **Live:** pampit.fun — Next.js, Rust, CosmWasm

Production token launchpad across 6 Cosmos chains with zero backend infrastructure. Custom RPC batching client reducing distributed query latency 10x (4000ms → 400ms). Responsive UX with client-side pub/sub polling and pagination. All business logic in Rust smart contracts.

Blockchain Infrastructure Primitives

Open source reusable smart contract libraries in Rust/CosmWasm: **CW-Factory** (github) — Multi-index factory with SQL-like on-chain querying via dual-map architecture — **CW-ACL** (github) — Hierarchical access control with delegation chains — **CW-Thread** (github) — On-chain forum with DAG storage and O(1) lookups

Education

Bachelor of Science, Computer Science — University of Central Florida (2009-2013)

Bachelor of Arts, Comparative Religion — University of Florida (2003-2007)